

TIA–FPGA – FPGA replacement for the Atari 2600 Television Interface Adaptor (C010444)

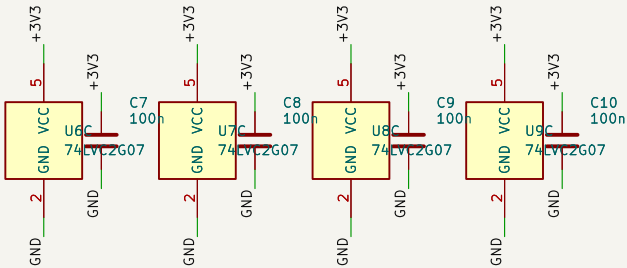
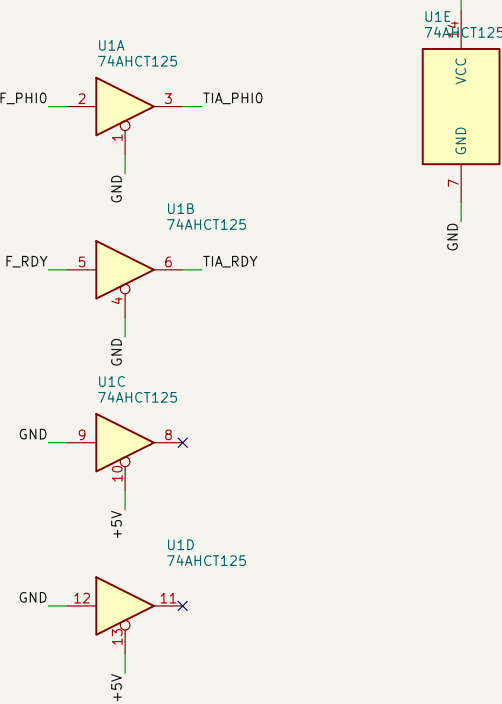
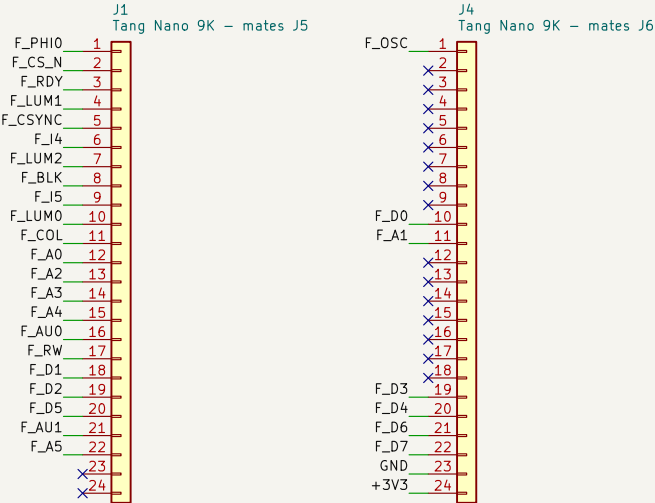
Rev B – digital interface with open–drain video and sound pads. The paddles do not fit the pin budget.

FPGA MODULE – Sipeed Tang Nano 9K (GW1NR–LV9QN88PC6/I5)

Pin map verified against both Sipeed board revisions (3672 and 3674), which agree. Of the 48 header pins only 29 are usable 3.3 V GPIO: 8 are BANK3 at 1.8 V, 8 are HDMI differential pairs, 3 are power. Rev B uses all 29, with none spare.

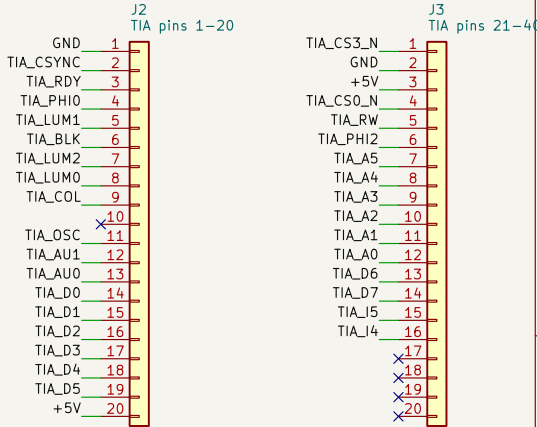
LEVEL TRANSLATORS

U1 is the critical part: the NMOS 6507 requires $V_{IH}(PHIO) = V_{cc}-0.2\text{ V} \sim 4.8\text{ V}$. A 3.3 V output will not clock the 6507 reliably.



TIA SOCKET – DIP–40

Two 20–way SIP strips that plug into the 2600 socket or a breadboard, in place of the C010444.



HDMI – not used by this design

The Tang Nano’s HDMI pins are dedicated differential pairs outside the 29–pin budget. A TMDS encoder in the FPGA could send colour out through them at no pin cost; the RTL has none yet.

FPGA pins (P/N): CK 69/68, D0 71/70, D1 73/72, D2 75/74.

RULE: leave J1/23–24 and J4/12–17 open – they carry the same TMDS pairs, and any stub degrades the signal.

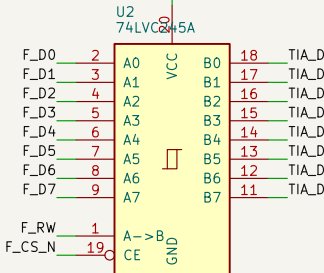
Pins 37–40 (paddles) and pin 10 (DEL, the colour trim) are left open: no FPGA pin is left for the paddles, and DEL has no use here.

J4/2–9 are BANK3 at 1.8 V – unusable at 3.3 V without a local LDO and a 1.8 V–powered buffer. Recovering those 8 pins is the way to the paddles.

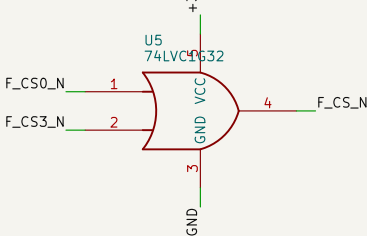
J4/18 (+5 V from the Tang Nano) left open on purpose: this board is fed from the breadboard rail, so tying both would put two sources on one net. Power the Tang Nano over USB. +3V3 is taken from J4/24.

U1 runs from +5 V – this is where the level step–up happens. Spare buffers 3 and 4: /OE tied to +5 V, input tied to GND.

U2 D0–D7 bidirectional. A side = FPGA, B side = TIA. DIR comes straight from the buffered R/W: DIR=1 (a read) sends FPGA → TIA. /OE comes from U5, so neither 245 control line costs an FPGA pin – that freed the one we needed.

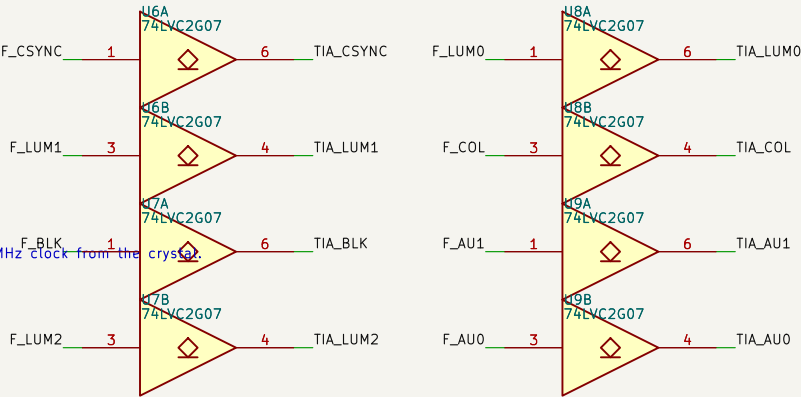


U5 /OE = /CS0 OR /CS3: the 245 drives the bus only while the TIA is selected. The FPGA reads the same net, F_CS_N, which frees the pin BLK uses.



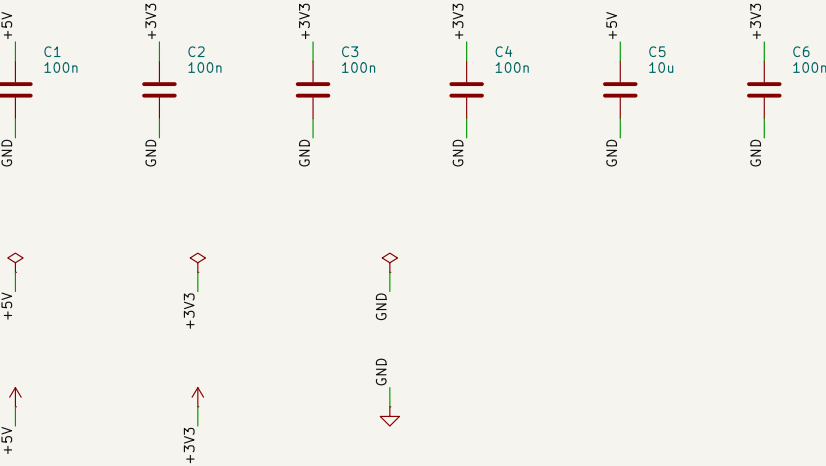
VIDEO AND SOUND PADS – U6–U9, 74LVC2G07: open drain, 5 V tolerant

On the die SYNC, LUM0–2, COL, BLK and both sound pads only ever pull down, so here the console’s own pull–ups set every level, as they do for the TIA. BLK pulls low through blanking: the 2600 and the 2600A from rev 16 feed it to the colour network.

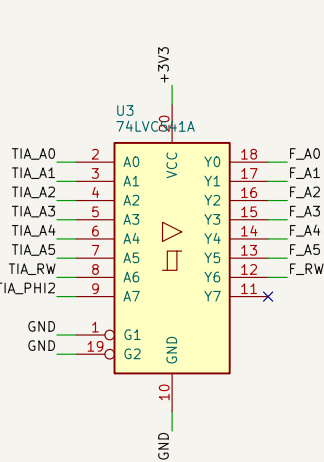


POWER AND DECOUPLING

+3V3 comes from the Tang Nano (J4/24). +5 V comes from the breadboard rail.



U3 A0–A5, R/W and PH2 – permanently enabled (G1=G2=GND).



U4 /CS0, /CS3, triggers and the 3.579545 MHz clock from the crystal.

